

# Math 152 Problem Set 2

Due 10/14/2026, at 10:00 PM

**Problem 1.** Use the  $\varepsilon, \delta$  definition of the limit to check that

$$\lim_{x \rightarrow 1} (2x^2 + 3) = 5$$

via the following steps. This will be similar to the continuity of  $f(x) = x^2$  in the lecture notes.

- (a) Write down the  $\varepsilon, \delta$  definition of the limit in this case.
- (b) Factor  $|(2x^2 + 3) - 5|$  as  $|x - 1| \cdot *$  for some expression  $*$ .
- (c) Show that if  $|x - 1| < 1$ , then  $* \leq C$  for some fixed number  $C$ .<sup>1</sup> (You should find a specific number here.)
- (d) Show that if  $|x - 1| < \min(1, \varepsilon/C)$ , then  $|(2x^2 + 3) - 5| < \varepsilon$ .

**Problem 2.** Consider the integral of the function  $f(x) = x$  over the interval  $[0, 1]$ . Find a partition  $\mathcal{P}$  of  $[0, 1]$  such that  $U(f, \mathcal{P}) - L(f, \mathcal{P}) < 1/5$ . You should compute the upper and lower sums explicitly. Drawing a picture may be helpful.

**Problem 3.** Suppose that a ball is thrown upward. We let  $y(t)$  denote the height of the ball at time  $t$ . We have the following information.

1. The ball is thrown from height 1, so  $y(0) = 1$ .
2. The ball has initial velocity 2 upward, so  $y'(0) = 2$ .
3. The ball is subject to a constant gravitational acceleration 1 pointing downward, so  $y''(t) = -1$  for all  $t$ .

With this information, answer the following questions:

- (a) Find a formula for the vertical velocity  $v(t) = y'(t)$  in terms of  $t$ .
- (b) Find a formula for  $y(t)$  in terms of  $t$ .
- (c) When does the ball hit the ground,  $y = 0$ ?
- (d) What is the ball's height at the point where  $v(t) = 0$ ?

**Problem 4.** Consider the parabola  $y = 8 - 2x^2$ .

- (a) Where is the graph of this parabola above the  $x$ -axis? Answer in the form of an interval.
- (b) Find the area between the parabola and the  $x$ -axis on this interval.

There are two optional problems on the next page. You are encouraged to think about these if you want more practice working with our definition of the integral.

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<sup>1</sup>You can use the *triangle inequality* here. This says that for any numbers  $a, b$ ,  $|a + b| \leq |a| + |b|$ .

In the following (optional) problem, you can use this fact.

**Proposition.** For any real numbers  $a, b$  with  $a < b$ , the interval  $[a, b]$  contains both rational and irrational numbers.

**Problem 5** (Optional). Consider the *Dirichlet function*:

$$D(x) = \begin{cases} 0 & x \text{ is irrational} \\ 1 & x \text{ is rational} \end{cases}$$

- (a) Explain in terms of the definition of the limit why the Dirichlet function is discontinuous at every point  $a \in \mathbb{R}$ .
- (b) Explain why for any partition  $\mathcal{P}$  of the interval  $[0, 1]$ , we have  $L(D, \mathcal{P}) = 0$  and  $U(D, \mathcal{P}) = 1$ . This means that these upper and lower bounds do not give us a unique value for the integral

$$\int_0^1 D(x) dx.$$

In our definition of the integral, we assume that the function is continuous, in which case this does not happen.

**Problem 6** (Optional). Suppose that  $f$  is a continuous function on an interval  $[a, b]$ , and  $c_1$  and  $c_2$  are constants such that  $c_1 \leq f \leq c_2$  on  $[a, b]$ . Argue directly from the definition of the integral<sup>2</sup> that

$$c_1(b-a) \leq \int_a^b f(x) dx \leq c_2(b-a)$$

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<sup>2</sup>Hint: Consider the one-piece partition  $\mathcal{P} = \{a < b\}$  of  $[a, b]$ . What are  $L(f, \mathcal{P})$  and  $U(f, \mathcal{P})$ ? Compare these to the integral and to  $c_1(b-a)$ ,  $c_2(b-a)$ .